Next-Generation Biometric Architecture for Classroom Environments: A Deep Analysis of Transformer-Based Recognition, Generative Enrollment, and High-Density Inference

1. Executive Summary

The implementation of automated attendance systems within educational institutions represents a distinct class of computer vision problems, characterized by a unique "semi-constrained" environment. Unlike high-security access control, where subjects are cooperative and processed individually, or city-scale surveillance, where the database is effectively infinite, the classroom presents a bounded but highly dynamic scenario: a known population (students), moderate density (50–200 subjects), and significant unconstrained variability in pose, occlusion, and appearance.

The baseline architecture presented—utilizing **YOLOv8-face** for detection, **ArcFace (Glint360k)** for recognition, **FAISS** for retrieval, **Streamlit** for visualization, and **MongoDB** for persistence—constitutes a competent "Generation 2.0" biometric stack. It leverages the strengths of widely adopted Convolutional Neural Networks (CNNs) and efficient vector search. However, the reported challenges regarding **batch inference latency**, **normalization consistency**, and **robustness to biological variance** (specifically shaving and partial occlusion) indicate that the system has reached the asymptotic performance limits of standard CNN-based pipelines.

This report provides an exhaustive analysis of the 2025 landscape for face recognition, specifically addressing the user's inquiry regarding the role of **Transformer architectures**. The analysis confirms that the shift from CNNs to **Vision Transformers (ViTs)**—exemplified by models like **LVFace** and **SwinFace**—is not merely an incremental upgrade but a fundamental architectural necessity for handling the "occlusion-heavy" nature of classroom environments. Furthermore, this report proposes a paradigm shift in enrollment methodology: moving from "Single-Shot Static Enrollment" to **Generative Gallery Augmentation (GGA)** using **InstantID**. By synthetically generating photorealistic variations of a student (e.g., bearded, looking down, varying lighting) from a single reference photo, the system can resolve the "shaving" and "pose" challenges at the database level, significantly reducing the burden on real-time inference models.

The recommended next-generation architecture integrates **SCRFD** for high-recall detection in

crowds, **LVFace** for occlusion-robust recognition, **Triton Inference Server** for dynamic high-throughput batching, and **FAISS FlatIP** for exact, zero-latency retrieval, forming a cohesive solution capable of real-time, high-accuracy attendance tracking in unconstrained academic settings.

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2. Technical Landscape Analysis: Deconstructing the Current Stack

To prescribe a robust path forward, it is essential to rigorously deconstruct the existing components of the user's stack (YOLOv8, ArcFace, FAISS) to understand why they falter under the specific constraints of a classroom.

2.1 The Physics of Classroom Surveillance

A classroom CCTV feed creates a specific "hostile" environment for computer vision, distinct from selfie-based verification:

1. **The Scale-Distance Ratio**: In a typical lecture hall, a camera mounted 3 meters high and 10 meters back results in students in the rear rows occupying as little as 15×15 pixels.
2. **The Occlusion Matrix**: Unlike a street scene where pedestrians move apart, seated students create "static occlusion" chains. A student in Row 2 permanently obscures 30% of the face of the student in Row 3.
3. **The One-Shot Domain Gap**: The enrollment photo is typically a high-quality, evenly lit ID card image (Domain A). The inference image is low-resolution, top-down, and shadowed (Domain B). This "Domain Gap" is the primary source of false negatives.

2.2 Face Detection: The Limitations of YOLOv8

YOLO (You Only Look Once) architectures are optimized for global object detection (cars, pedestrians). While yolov8-face is a popular adaptation, it inherits design choices that are suboptimal for dense face detection.

- **Anchor Stride and Small Faces**: YOLO models typically use a stride of 8, 16, or 32 pixels. A stride of 32 means the model processes the image in 32×32 grids. If a face is smaller than this grid (e.g., 20 pixels), the feature map loses the spatial granularity required to distinguish the face from the background or a neighbor's shoulder.[1]
- **IoU-Based Loss**: YOLO relies on Intersection over Union (IoU) for bounding box regression. In a crowded classroom, faces often overlap significantly (e.g., one head behind another). High IoU thresholds in Non-Maximum Suppression (NMS) can suppress the valid detection of a partially occluded student, marking them as "background".[2]
- **Performance vs. Recall**: While YOLOv8 is fast (high FPS), in an attendance system,

Recall is paramount. Missing a student (False Negative) is worse than a slight delay. Benchmarks on the **WIDER FACE** dataset (Hard subset) consistently show that specialized detectors like **SCRFD** or **RetinaFace** outperform YOLO variants by 10–15% in Average Precision (AP) for small faces.[1]

2.3 Face Recognition: The CNN Receptive Field Bottleneck

ArcFace (ResNet-100) is the industry standard for 2020–2023. It utilizes a Residual Network (CNN) backbone trained with Additive Angular Margin Loss.

- **Local Receptive Fields**: CNNs build representations from the bottom up. Convolutions aggregate local pixel patterns (edges $\rightarrow$ textures $\rightarrow$ features). This creates a "local bias."
- **The Occlusion Failure Mode**: If a student wears a mask or rests their chin on their hand, the local features of the lower face are corrupted. A CNN, lacking a global attention mechanism, struggles to "dynamically ignore" these corrupted regions. It propagates the "mask texture" into the final embedding, pushing the vector away from the "clean face" cluster in the hypersphere. This results in the system failing to match the student, or worse, matching them to a generic "unknown" cluster.[4]
- **The Shaving Challenge**: The user specifically notes "changes like shaving." A beard dramatically alters the texture and geometry of the lower face. For a CNN trained primarily on structural features, a beard is a massive occlusion. Unless the training dataset (Glint360k) contains the *specific* individual with and without a beard, the model's intra-class variance often fails to bridge the gap between "Bearded Student A" and "Clean-Shaven Student A".[6]

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3. Face Detection in Crowded Environments: The Shift to Specialized Solvers

To address the "small face" and "batch inference" challenges, the architecture must migrate from general-purpose detectors to specialized biometric solvers.

3.1 SCRFD: Sample and Computation Redistribution

SCRFD (Sample and Computation Redistribution for Face Detection) represents the current state-of-the-art for efficient, high-accuracy face detection, specifically designed to address the inefficiencies of models like YOLO and RetinaFace.

- **Mechanism**: SCRFD optimizes the "effective receptive field" of the network. It analyzes the distribution of face scales in the training data and redistributes the computational budget. It allocates more capacity to shallow layers (where small faces are detected) and less to deep layers (where large faces are detected), drastically improving the recall for the "back row" students without increasing total inference time.[1]

- **Anchor-Free Design**: Unlike YOLO's anchor boxes (which require heuristic tuning), SCRFD uses point-based regression. This is far more robust to the extreme aspect ratios found in crowded scenes (e.g., a partially hidden face that is wide but short).
- **Benchmark Performance**: On the WIDER FACE Hard dataset (representing crowds), SCRFD-10G achieves ~95% AP, compared to ~82% for standard YOLO adaptations.
- **Batching Advantage**: SCRFD is highly optimized for batch processing. When integrated with **Triton Inference Server**, it can process a batch of 8–16 classroom frames simultaneously, outputting landmarks and bounding boxes in a single forward pass.

3.2 RetinaFace: The Dense Regression Alternative

RetinaFace is another strong contender, utilizing a Feature Pyramid Network (FPN) with a context head module.

- **Dense Localization**: RetinaFace performs pixel-wise face localization. It is exceptionally good at finding faces in "extreme" poses (e.g., a student looking down at a desk).
- **Landmark Accuracy**: It outputs 5 facial landmarks (eyes, nose, mouth corners) with higher precision than YOLOv8-face. These landmarks are critical for the **Normalization** step (alignment), directly impacting the recognition accuracy.

Recommendation: For a classroom setting, **SCRFD-10G** is the optimal choice. It offers the best trade-off between the high recall required for attendance and the low latency required for real-time tracking.[3]

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4. The Transformer Revolution: Exploring ViTs for Robust Recognition

The user asks: *Explore if Transformer architectures can play a role.*

The definitive answer is Yes. The transition from CNNs to Vision Transformers (ViTs) in 2024–2025 is driven precisely by the need to handle the "unconstrained" variables (occlusion, pose, expression) that confound traditional models.

4.1 Theoretical Advantage: Global vs. Local Attention

The fundamental difference lies in how the models process information:

- **CNN (ResNet/ArcFace)**: Fixed weights. The "convolutional kernel" looks at the image the same way regardless of the content. It aggregates neighborhood pixels. If the neighborhood is a "mask," it learns "mask features."
- **Transformer (ViT)**: Dynamic weights. The image is split into patches (tokens). The **Self-Attention** mechanism calculates the relationship between *every* patch pair.
  - *Scenario*: A student has a beard and glasses.
  - *ViT Response*: The self-attention matrix calculates that the "beard patches" are

inconsistent with the "forehead patches" or have low semantic clarity. The model dynamically assigns *lower attention scores* to the beard regions and *higher attention scores* to the periocular (eye) regions.
    - *Result*: The final embedding is constructed primarily from the reliable features (eyes, forehead), effectively "ignoring" the noise. This is not a heuristic; it is an emergent property of the attention mechanism.[4]

4.2 LVFace: The Large Vision Model for Face Recognition

LVFace is a landmark 2025 architecture that successfully adapts Large Vision Models (LVMs) to the specific geometric constraints of face recognition.

- **Progressive Cluster Optimization (PCO)**: Training ViTs for face recognition is notoriously difficult because they require massive datasets to converge. LVFace introduces PCO, a multi-stage training curriculum:
    1. **Feature Alignment**: Stabilizes the random initialization using a relaxed loss.
    2. **Centroid Stabilization**: Anchors the "identity centers" in the embedding space.
    3. **Boundary Refinement**: Tightens the clusters to maximize discriminative power.
- **Performance**: LVFace achieves State-of-the-Art (SOTA) results on **WebFace42M** (42 million images) and, crucially, won the **ICCV 2021 Masked Face Recognition Challenge** (continuing into 2025 benchmarks). This proven capability on masked/occluded faces makes it the ideal candidate for the user's classroom requirements.[4]
- **Architecture**: It typically uses a ViT-Base or ViT-Large backbone. While computationally heavier than ResNet-100, the accuracy gain in "hard" scenarios (like the user's single-photo constraint) justifies the cost.

4.3 SwinFace: The Hierarchical Hybrid

SwinFace utilizes the **Swin Transformer** (Hierarchical Vision Transformer using Shifted Windows).

- **Shifted Window Attention**: Unlike standard ViTs that compute global attention (quadratic complexity), Swin computes attention within local windows and "shifts" them to create cross-window connections.
    - *Benefit*: This creates a hierarchical understanding (Edges $\rightarrow$ Parts $\rightarrow$ Whole) similar to CNNs but with the adaptive weighting of Transformers.
- **Small Face Robustness**: Because it maintains a local window structure, SwinFace is often better at preserving the details of *small faces* (like those in the back of a class) compared to pure ViTs, which can sometimes "over-smooth" low-resolution features.
- **Multi-Task Synergy**: SwinFace is explicitly designed to handle multiple tasks (Recognition, Expression, Age, Attribute) in a single pass. This allows the system to not only mark attendance but potentially analyze "Student Engagement" (via expression/pose) without running a second model.[11]

4.4 FaceXFormer: The Unified Part-Based Approach

FaceXFormer treats different facial analysis tasks as "tokens" within a single unified framework.

- **Occlusion Handling**: It is highly effective for "Part-based" recognition. By implicitly parsing the face (understanding where the nose, eyes, and mouth are), it can selectively suppress occluded parts.
- **Relevance**: While powerful, it is currently more complex to deploy than LVFace. It is a strong candidate for future research but LVFace is more "production-ready" for immediate deployment.[13]

Conclusion on Transformers: For the user's specific need to handle **partial occlusion** and **changes like shaving**, **LVFace** is the superior architectural choice. Its attention mechanism provides the necessary plasticity to map a "bearded face" and a "clean-shaven face" to the same identity cluster, provided the upper facial features remain consistent.

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5. Solving the "Single Photo" Constraint: Generative Enrollment Pipeline

The requirement to "detect students from a single database photo" while handling biological changes is the single greatest point of failure in current systems. Mathematical projections show that the **intra-class variance** (difference between the same person's photos) often exceeds **inter-class variance** (difference between two people) when conditions change drastically (e.g., severe lighting change + beard). A single vector point is insufficient to define the "manifold" of a person's identity.

Recommendation: Instead of relying on the recognition model's invariance, we must **Augment the Gallery** using Generative AI. We call this **Generative Gallery Augmentation (GGA)**.

5.1 The InstantID Mechanism

InstantID (2024/2025) is a breakthrough in zero-shot identity-preserving image generation. It decouples the "Style" and "Content" of an image from the "Identity".[14]

- **Pipeline**:
  1. **IdentityNet (ControlNet)**: Extracts the strong semantic identity features from the reference face (the single database photo).
  2. **IP-Adapter**: Acts as a "prompt" for the diffusion model (e.g., SDXL), injecting the facial features into the generation process.
  3. **Diffusion Generation**: The model generates a *new* image based on a text prompt ("a

man with a beard") while rigidly constraining the pixel generation to match the identity features from the reference.

5.2 Implementation: The "Multi-Shot" Enrollment Strategy

When a new student is enrolled with a single ID card photo, the system should triggers a background batch job:

1. **Input**: Student_ID_123.jpg (Clean shaven, frontal).
2. **Generation Loop**: The system feeds this image to InstantID with specific prompts:
   - *Prompt A*: "A photo of a person with a full beard, photorealistic, 8k."
   - *Prompt B*: "A photo of a person looking down at a book, side profile."
   - *Prompt C*: "A photo of a person wearing reading glasses."
   - *Prompt D*: "A photo of a person in low light, shadowy environment."
3. **Vector Extraction**: The system passes these 4 synthetic images + the 1 original image through the Recognition Model (LVFace).
4. **Gallery Indexing**: The system stores **5 vectors** for Student_ID_123 in the FAISS index.

5.3 The Mathematical Impact on Inference

- **Scenario**: The student comes to class with a beard.
- **Without GGA**: The live vector V_{live} is compared to the single reference V_{ref}. Due to the beard, Cosine Similarity $\cos(\theta) = 0.35$. **Result**: Unknown (False Negative).
- **With GGA**: The live vector V_{live} is compared to the set $\{V_{ref}, V_{beard}, V_{profile}, \dots\}$.
  - $\cos(V_{live}, V_{ref}) = 0.35$
  - $\cos(V_{live}, V_{beard}) = 0.78$
- **Result**: Match found. The system successfully identifies the student despite the biological change.
- **Privacy Note**: The synthetic images are **transient**. They are generated, vectorized, and immediately deleted. Only the mathematical embeddings are stored, ensuring compliance with privacy standards regarding biometric data retention.[16]

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6. Vector Search and Database Optimization

The user employs **FAISS** (Facebook AI Similarity Search). However, for a classroom setting, the standard configuration (usually IndexHNSW or IndexIVF) is often suboptimal.

6.1 The Fallacy of Approximate Search (ANN)

Algorithms like **HNSW (Hierarchical Navigable Small World)** are designed for "Approximate Nearest Neighbor" search. They sacrifice a small amount of accuracy (Recall) for massive

speed gains, typically scaling to millions or billions of vectors.

- **Classroom Scale**: A typical attendance database has $N < 1,000$ identities. Even with Generative Enrollment (5 vectors per person), $N < 5,000$.
- **Performance Reality**: At $N = 5,000$, a brute-force exact search takes microseconds on a modern CPU. The overhead of traversing the HNSW graph structure may actually be *slower* than a simple matrix multiplication, and it introduces a non-zero risk of missing the best match.[18]

6.2 Recommendation: FAISS IndexFlatIP

We recommend using **IndexFlatIP** (Inner Product) with strict normalization.

- **Mechanism**: This index performs an exact dot product between the query vector and every vector in the database.
- **Normalization**:
  - The embeddings from LVFace (or ArcFace) must be L2-normalized ($||v|| = 1$).
  - Mathematically, for normalized vectors, the Dot Product equals Cosine Similarity: $A \cdot B = ||A||||B|| \cos(\theta) = 1 \cdot 1 \cdot \cos(\theta)$.
- **Implementation**:
  Python
  ```python
  # Python FAISS Example
  import faiss
  import numpy as np

  # Dimension of LVFace/ArcFace embeddings
  d = 512

  # Create Exact Index
  index = faiss.IndexFlatIP(d)

  # Add Vectors (Must be float32 and Normalized)
  index.add(database_vectors)

  # Search
  D, I = index.search(query_vectors, k=1)
  ```
- **Benefit**: This guarantees **100% Recall**. The system will *always* find the closest vector in the database, eliminating "search error" as a source of failure.

6.3 Database Architecture: Redis + MongoDB

While MongoDB is excellent for persistent storage (logs, student profiles), it is too slow for real-time state management.

- **Redis (In-Memory)**: Use Redis to store the "Active Classroom State."
  - *Key*: Room_101_Attendance
  - *Value*: List of detected IDs in the last 5 seconds.
- **Tracking Logic**: The Streamlit dashboard should poll Redis, not MongoDB. MongoDB should be updated asynchronously (e.g., once per minute) to finalize the attendance records. This decoupling prevents database I/O from blocking the inference loop.

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7. System Architecture and Deployment Strategy

To address the "Batch Inference" and "Latency" challenges, the system must move away from a monolithic Python script (Streamlit) to a microservices architecture.

7.1 The Bottleneck of Python Inference

Running deep learning models directly inside a Streamlit loop is inefficient. Python's Global Interpreter Lock (GIL) prevents true parallelism. Processing 50 faces sequentially in a loop will result in < 5 FPS.

7.2 Solution: Triton Inference Server

NVIDIA Triton Inference Server is the industry standard for production AI.

- **Dynamic Batching**: This is the critical feature for classrooms.
  - *Scenario*: The detector finds 40 faces in Frame 1.
  - *Without Triton*: The Python script loops 40 times.
  - *With Triton*: The script sends 40 requests to Triton. Triton's "Dynamic Batcher" aggregates them into a single tensor of shape `` and fires the GPU *once*.
  - *Impact*: Massive throughput increase. The GPU is saturated efficiently rather than waiting for Python overhead.[20]
- **Concurrent Execution**: Triton can run the Detection Model (SCRFD) and Recognition Model (LVFace) on the same GPU simultaneously, managing memory allocation automatically.

7.3 Multi-Object Tracking (MOT): ByteTrack

To handle "Partial Occlusion" and "Unknowns" robustly, we cannot rely on single-frame detection. We must track identities over time.

- **ByteTrack**: We recommend **ByteTrack** over DeepSORT.
  - *Why?*: DeepSORT discards low-confidence detections (e.g., a face partially hidden by a hand). ByteTrack utilizes a two-stage association process. It first matches high-confidence detections, then associates the remaining low-confidence

detections to existing tracks.
    - *Classroom Benefit*: If a student turns their head or covers their mouth, the detection confidence drops. ByteTrack keeps the ID alive during this "low confidence" period, preventing the system from re-assigning a new "Unknown" ID when the student looks back at the camera.[22]

7.4 Architecture Diagram (Text-Based)

1. **Input Layer**: 4K IP Camera (RTSP Stream).
2. **Preprocessing Service**: Decodes video, resizes frame to 1280×1280 (for SCRFD).
3. **Inference Layer (Triton)**:
    - **Model A**: SCRFD (Outputs: BBoxes + Landmarks).
    - **Tracker**: ByteTrack (Assigns Track_ID).
    - **Logic**: Selects the "Best Frame" for each Track_ID (based on size/centrality) every 2 seconds.
    - **Model B**: LVFace (Outputs: 512-d Embedding).
4. **Matching Layer**:
    - FAISS IndexFlatIP (Queries embeddings against Generative Gallery).
    - **Voting System**: A Track_ID is confirmed as Student_X only if 3 out of 5 queries match Student_X.
5. **State Layer**: Redis (Stores Track_ID $\leftrightarrow$ Student_Name map).
6. **Presentation Layer**: Streamlit (Visualizes bounding boxes and names from Redis).

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8. Robustness Strategies: Unknowns, Normalization, and Ethics

8.1 Handling "Unknowns": Open-Set Recognition

In a closed-set (1:N) system, every probe is assumed to be in the gallery. In a classroom, "Unknowns" (intruders or unenrolled students) exist.

- **The Threshold Problem**: A fixed threshold (e.g., 0.5) is brittle.
- **Quality-Aware Thresholding**:
    - Use a quality metric (e.g., MagFace Quality Score or simply the face resolution).
    - **Formula**: $T_{dynamic} = T_{base} + \lambda(1 - Q_{score})$.
    - *Logic*: If the face image is blurry (low quality), we demand a *higher* similarity score to confirm identity. If the face is crystal clear, we accept a slightly lower score. This reduces False Positives on blurry background faces.
- **Spatio-Temporal Constraints**: If Face #5 is identified as "John" with 90% confidence in Frame 100, and then in Frame 101 the confidence drops to 40% (due to occlusion), the system should NOT label Frame 101 as "Unknown." It should inherit the label "John" from

the Tracker history. This "Temporal Smoothing" is the most effective way to eliminate flickering "Unknown" labels.[5]

8.2 Normalization: The Art of Alignment

"Normalization" in the user query likely refers to **Face Alignment**.

- **Standard**: ArcFace/LVFace require a specific 5-point alignment (similarity transformation) to standard coordinates.
- **The Trap**: Many users just crop the bounding box. This destroys accuracy because the eyes are not aligned to the neurons expecting them in specific positions.
- **Requirement**: The system *must* use the 5 landmarks from SCRFD to perform an affine transformation (rotation + scaling) to align the eyes horizontally and center the nose, producing a standard 112×112 input tensor.
- **Illumination Normalization**: Do not use Histogram Equalization (HE) on RGB images; it distorts colors. Use **CLAHE** (Contrast Limited Adaptive Histogram Equalization) on the **L-channel** of the LAB color space if the classroom has harsh lighting (e.g., sun glare).

8.3 Ethical and Privacy Considerations

While technical, the deployment of this system involves sensitive biometric data.

- **Data Minimization**: The system should store *only* the embeddings (vectors), not the face chips, in the long-term database.
- **Transient Generation**: The synthetic images generated by InstantID must be treated as temporary buffers and deleted immediately after enrollment.
- **Transparency**: Students should be informed that the system uses "Identity-Preserving Generative AI" to improve accuracy, ensuring transparency about the use of their data.

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9. Hardware Recommendations

To support this high-density architecture (LVFace + Triton + InstantID Enrollment):

Component	Minimum Spec (Edge)	Recommended Spec (Server)	Reasoning
GPU	NVIDIA Jetson Orin NX (16GB)	NVIDIA RTX 4070 or A4000 (16GB+)	ViT models (LVFace) and InstantID (Enrollment) are VRAM intensive.

CPU	ARM Cortex-A78AE	AMD Ryzen 9 / Intel Core i9	Preprocessing (decoding 4K video) and ByteTrack logic run on CPU.
RAM	16 GB	64 GB	FAISS and Docker containers (Triton) require substantial overhead.

Cloud vs. Edge: For a single classroom, an Edge device (Jetson) is sufficient *if* enrollment (InstantID) is offloaded to a cloud/desktop server. InstantID is too heavy to run efficiently on a Jetson for real-time enrollment. The recommended split is: **Enrollment on Cloud/Desktop GPU**, **Inference on Edge Device**.

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10. Conclusion

The transition from the user's current **YOLO/ArcFace** stack to the proposed **SCRFD/LVFace/InstantID** architecture represents a leap from "generic computer vision" to "domain-adaptive biometrics."

1. **Transformers are Essential**: **LVFace** provides the global attention mechanism necessary to "see through" the occlusions (masks, hands, phones) that define the classroom environment, solving the robustness issue where CNNs fail.
2. **Generative Enrollment is the Key Enabler**: The "Single Photo" constraint is solved not by better recognition, but by better data. **InstantID** allows the system to mathematically anticipate biological changes (shaving) before they happen in the real world.
3. **Specialization Wins**: Moving from general object detectors (YOLO) to specialized face solvers (SCRFD) and from approximate search (HNSW) to exact search (FlatIP) optimizes the system for the specific physics of the classroom.

By implementing this architecture, the attendance system will evolve from a fragile prototype to a resilient, professional-grade solution capable of handling the messy, dynamic reality of student life.

References (Integrated Context)

- **Detection**: Comparison of YOLO vs. SCRFD/RetinaFace performance on crowded datasets.[1]
- **Transformers**: LVFace architecture and PCO training [4]; SwinFace hierarchical attention.[11]

- **Enrollment**: InstantID identity-preserving generation [14]; Facial hair impact on recognition.[6]
- **Inference**: Triton Inference Server architecture [20]; FAISS indexing strategies.[19]
- **Tracking**: ByteTrack vs. DeepSORT performance.[22]

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